

RESEARCH INTENT

WQD7002/7023/7025 Data Science Research Project

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Area of Research	Biomedical Engineering and Medical Image Analysis
Research Title	Smart Wound Monitoring: AI-Driven Image Analysis Based on the T.I.M.E. Framework
Introduction	Many of the patients worldwide are suffering from chronic wounds, which often lead to serious infection, and may be life threatening if it is not monitored properly. Wound assessment still relies heavily on clinicians' observation, which is very subjective and time-consuming. Therefore, there is a need to develop an AI-driven wound assessment system that evaluates wound condition based on the T.I.M.E. framework and provides appropriate recommendations to support both clinicians and patients.
Statement of Aim	This research aims to develop a smart wound monitoring system that utilizes AI to analyze wound images based on the T.I.M.E. framework and provides some recommendations for appropriate treatment or dressing options to improve the wound healing process.
Research Questions	1. How can AI models be trained to evaluate the wound conditions based on T.I.M.E. criteria using clinical image? 2. What is the relevant image-based features that indicate each T.I.M.E. components? 3. How does the system integrate this analysis into a mobile application that supports both clinician supervision and patient self-monitoring?
Research Objectives	1. To develop AI models that classify and evaluate wound condition based on the T.I.M.E. criteria using image-based features. 2. To integrate the wound assessment model into a mobile application for patients and clinicians, to enable real-time monitoring and personalized care recommendations. 3. To validate the accuracy of the system through testing with clinical dataset and clinician feedback.
Scope	This research focuses on image based wound assessment to analyze tissue type, the presence of infection, and the overall healing progress.

Significance of the Research	By incorporating the T.I.M.E. framework into an AI-driven system, this research aims to improve clinical accuracy, promotes highly efficiency in wound monitoring and wound healing process. It can help in improving healthcare accessibility and early detection of wound deterioration.
Methodology	This research will be using publicity available dataset to perform image preprocessing, model training for T.I.M.E. classification, and validation against held-out data with standard metrics.
Expected Outcomes	A functional AI-assisted mobile application that can analyze wound images based on T.I.M.E. criteria.
Skills Needed	Deep learning, Image processing, Mobile app development, Data processing, Model evaluation, Performance tuning, Wound care knowledges.
Timeline/ Gantt Chart	Week 1 – 3: Research on the desired project topics Week 4 – 6: Literature review, draft methodology, and confirming the dataset used Week 7 – 9: Baseline model prototyping Week 10 – 12: Draft proposal, prepare presentation slides Week 13 – 17: Proposal Presentation


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